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Facilitator Guide

Sector

Green Jobs

Sub Sector

Green Hydrogen

Occupation

Entrepreneur

Reference Id

SGJ/Q0121, Version 2.0 NSQF Level - 5

Green Hydrogen Plant Entrepreneur

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development then Skill Development
should be our mission."

Shri Narendra Modi
Prime Minister of India

Certificate

COMPLIANCE TO
QUALIFICATION PACK – NATIONAL OCCUPATIONAL STANDARDS

is hereby issued by the

SKILL COUNCIL FOR GREEN JOBS

for the

SKILLING CONTENT: FACLITATOR GUIDE

Complying to National Occupational Standards of
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Acknowledgements

This “Facilitator Guide for Green Hydrogen Plant – Entrepreneur” is dedicated to all aspiring youth who desire to gain special skills and gain a meaningful livelihood in the Green Hydrogen sector.

About this Book

This Facilitator Guide is designed to enable training for the specific Qualification SGJ/Q0121: Green Hydrogen Plant Entrepreneur.

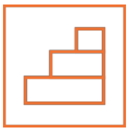
“To make India the Global Hub for production, usage and export of Green Hydrogen and its derivatives. This will contribute to India’s aim to become Aatmanirbhar through clean energy and serve as an inspiration for the global Clean Energy Transition. The Mission will lead to significant decarbonisation of the economy, reduced dependence on fossil fuel imports, and enable India to assume technology and market leadership in Green Hydrogen.”

Government of India is aiming towards developing a capacity and solution to produce 5MMT of Green Hydrogen by 2030 and 60-100Gw Electrolyzer installations and 125Gw Renewable energy production for Green Hydrogen and 50MMT of Carbon abatement cumulatively and creation of 6 lakh new green jobs.

The contents of this book are in simple language, without going into too much theoretical details and calculations. It is envisaged that this training manual will provide the participants with the knowledge and skills required for the Green Hydrogen Plant Entrepreneurship complying with all applicable codes, standards, and safety requirements. Skill Council for Green Jobs has designed and implemented a range of skilling interventions with a focus on Project construction, system Installation and Operations & Maintenance related job roles for Green Hydrogen. This updated Participant book is designed to enable theoretical and practical training on Green Hydrogen plant Entrepreneur, SGJ/Q0121 and is available for free download. The contents of this book are in simple language, without going into too much theoretical details and calculations. It is envisaged that this training manual will provide the participants with the knowledge and skills required for Green Hydrogen Plant Entrepreneur.

Symbols Used

Key Learning Outcomes



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Unit 1.9 - Role and Responsibilities of a Green Hydrogen Plant Technician

SGJ/N1817

Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss the properties and characteristics of Hydrogen
2. Discuss various colour code nomenclature of Hydrogen and role of Green Hydrogen in sustainable energy transition.
3. Discuss key aspects related to production, storage and transportation of Green Hydrogen.
4. Discuss key aspects of the hydrogen economy
5. Discuss the applications of Green hydrogen in industry, transport and power production.
6. Discuss in brief the emerging entrepreneurial opportunities across Green Hydrogen Value chain
7. Identify opportunities to various renewable energy sources
8. Discuss the role and responsibilities of a Green Hydrogen Plant Entrepreneur.

UNIT 1.1: Necessity Of Green Hydrogen in Sustainable Energy Transition

Unit Objectives

At the end of this unit, you will be able to:

1. Describe the concept of sustainable energy
2. Explain the necessity of green hydrogen in sustainable energy transition

Ask

Ask the participants if they know the meaning of the term 'green energy'.

Notes for Facilitation

- Start by giving an introduction about green energy.
- Tell them that green energy is also known as renewable energy as it is produced from natural resources such as sun, wind or water.
- Tell them about the importance of shifting from conventional, fossil fuel-based energy systems to sustainable and environmentally friendly sources of energy.
- Tell them that Green Hydrogen is one such source of renewable energy.
- Tell them about the negative impact the growth of fossil fuel-based electricity generation has on environment.
- Explain that Green Hydrogen can be used:
    - For blending with piped natural gas (PNG)
    - In domestic, commercial, and industrial sectors
    - To power heavy-duty long-haul transportation vehicles
    - To store energy
    - In combination with renewable energy sources to ensure a stable power supply
- List the various reasons for considering green hydrogen for substantial energy transition, such as:

Fig. 1.1.1: Reasons for Considering Green Hydrogen for Substantial Energy Transition

UNIT 1.2: Properties, Characteristics and Nomenclature of Hydrogen

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the properties and characteristics of Hydrogen.
2. Describe basic concepts of Hydrogen as an energy carrier.
3. Discuss various colour code nomenclature of Hydrogen.

Ask

Ask the participants if they know the properties and characteristics of Hydrogen.

Notes for Facilitation

- Start by giving an introduction about the atomic structure of hydrogen.
- Tell them that the arrangements of components in an atom is known as its atomic structure.
- Tell them that the components of an atom are:
    - Electrons
    - Protons
    - Neutrons
- Draw a diagram of the atomic structure of any atom, say carbon and explain its atomic structure.
- List the various properties of hydrogen tell them that hydrogen is:

Colourless	Odourless	Tasteless	Light
Non-toxic	Combustible	Insoluble in water	

Fig. 1.2.1: Properties of hydrogen

- Tell them about the atomic structure of hydrogen.
- Tell them about the isotopes of hydrogen, which are as follows:
    - Protium

- Deuterium
- Tritium

- Explain with the help of diagrams of the atomic structures of all the three isotopes.
- Tell them that since hydrogen is a very light gas, it can be compressed and takes very less storage space.
- Say that, to understand why hydrogen is a better alternative to diesel engine and fuel cell, let us understand the concept of diesel engine and fuel cell.
- Tell them that a fuel cell is similar to a battery and converts chemical energy not electrical energy.
- Explain with the help of a diagram of a fuel cell.
- Draw a comparison between diesel driven internal combustion engine and the fuel cell and between hydrogen and other fuels.
- Tell them that the primary factor behind shifting to hydrogen for energy is that hydrogen carries over twice the energy per unit of weight compared to natural gas and diesel.

Ask

Ask the participants if they can tell some uses of hydrogen as an energy carrier.

Notes for Facilitation

- List some uses of hydrogen as an energy carrier.
- Tell them that hydrogen can be produced from different sources using different processes.
- Based on the source, different colour codes have been assigned to it. This makes it easier to identify the source of hydrogen.
- Tell them that the colour codes are as follows:

Grey hydrogen	Blue hydrogen	Green hydrogen	Black hydrogen
Pink hydrogen	Turquoise hydrogen	Yellow hydrogen	

Fig. 1.2.2: Colours of hydrogen

- Explain the colour coding of hydrogen in detail.
- Explain the process of producing hydrogen based on the colour code.

- Solution to Activity: Match the process and source of production of hydrogen with the colour codes of hydrogen
    - A-H, B-G, C-E, D-F

- Solution to Activity: Process and the source of input energy

Hydrogen colour Code	Process	Source of input energy
Grey	SMR	Natural gas
Blue	SMR	Natural gas
Green	Electrolysis	Renewable energy
Black	Gasification	Coal
Pink	Electrolysis	Nuclear energy
Turquoise	Pyrolysis	Natural gas
Yellow	Electrolysis	Solar energy

UNIT 1.3: Value Chain for Green Hydrogen

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the concept of value chain for Green Hydrogen.
2. Explain the elements of a value chain of green hydrogen.
3. Discuss benefits and drawbacks of existing methods of Hydrogen production.

Ask

Ask the participants if they know they know the concept of value chain.

Notes for Facilitation

- Start by explaining the concept of value chain.
- Tell them that the value chain of a product can be depicted as follows:

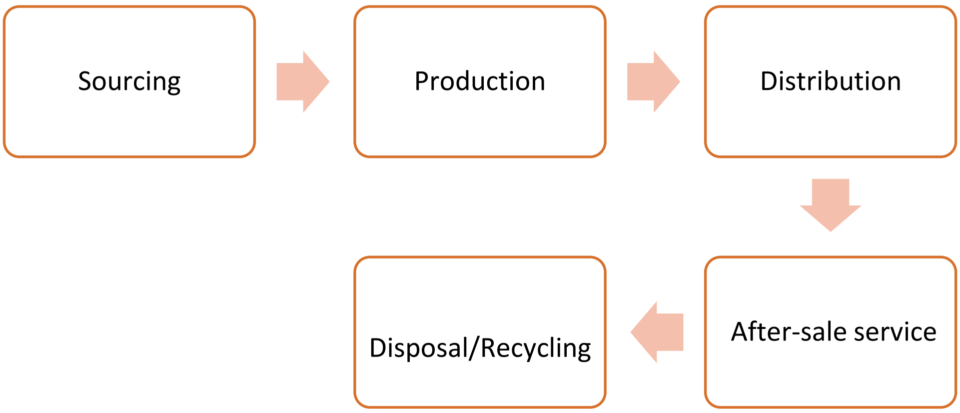


Fig.1.3.1: Value chain of a product

- Tell them that the value chain of hydrogen consists of the following three stages:

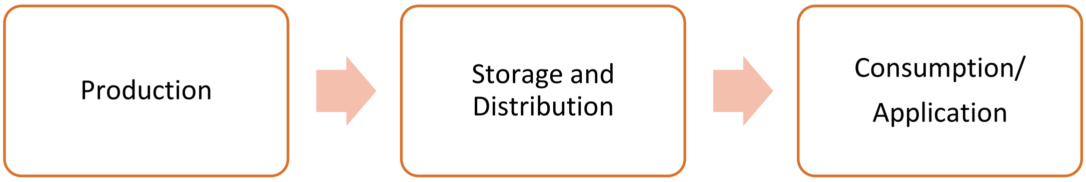


Fig.1.3.2: Value chain of hydrogen

- Explain the value chain of hydrogen in detail.
- Tell them about the benefits and drawbacks of the different hydrogen production methods, such as:
    - Steam methane reformation
    - Coal gasification
    - Pyrolysis
    - Biomass gasification
    - Electrolysis
- Tell them about the three different type of electrolysis methods, such as:
    - AEL
    - PEM
    - SOEL
- Tell them about the benefits and drawbacks of the different electrolysis methods.

UNIT 1.4: Key Aspects and Challenges related to Production, Storage, Transportation, and Distribution of Green Hydrogen

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the challenges in the production and storage of green hydrogen
2. Explain the challenges in the transportation and distribution of green hydrogen

Ask

Ask the participants if they know they can tell some challenges faced in the storage and production of green hydrogen.

Notes for Facilitation

- Tell them that the main challenge faced during production of hydrogen is the cost of production technologies.
- Tell them about the initiative taken by the Indian government to provide incentives for producing hydrogen.
- Tell them that it is cheaper to store fossil fuels, as:

1. •They can be stored at regular atmospheric pressure
2. •They are not inflammable.
3. •They occupy less space.

Fig.1.4.1: Storing fossil fuels

- Tell them that, in comparison to fossil fuels, storing hydrogen becomes expensive because of the following reasons:

- Low Energy Density
- Lekage Concerns
- Embrittlement of Materials
- Storage Methods
- Safety Considetations
- Infrastructure Devlopment

Fig.1.4.2: Storing hydrogen

- Tell them that the following challenges are faced while transporting hydrogen:

Low Energy Density	Safety	Distance and Efficeincy
Cost	Material Compatibility	Storage Duing Transit

Fig.1.4.3: Transporting hydrogen

- Tell them about the challenges faced while distributing hydrogen.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank papers and pens to each group.
- Ask the groups to make three columns on paper and list the challenges faced in storing, transporting and distributing hydrogen.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 1.5: End Use Applications of Green hydrogen in Industry, Transport and Power Production

Unit Objectives

At the end of this unit, you will be able to:

1. Understand the applications of green hydrogen in industry and transport sectors
2. Explain the application of green hydrogen in electric power generation

Ask

Ask the participants if they know they can tell some uses of green hydrogen.

Notes for Facilitation

- Tell them that the main use of green hydrogen is as fuel, for heating and as feedstock.
- Tell them about the uses of green hydrogen across industries.
- Tell them that, in agriculture industry, green hydrogen can be used as follows:

Power machinery and equipment	Tractors and irrigation systems
Converted into ammonia	Component of fertilizers

Fig.1.5.1: Uses of hydrogen in agriculture sector

- Tell them that, in petroleum refining industry, green hydrogen can be used as follows:

Petroleum refining	Ammonia production
Methanol production	Other uses

Fig.1.5.2: Uses of hydrogen in petroleum refining industry

- Tell them that green hydrogen can be used in other industry, such as:

Iron and steel	As a fuel source, in place of natural gas
Food	To convert unsaturated fats into to saturated oils and fats
	To extend shelf life of packaged food
	In food additives
Metal works	In metal alloying and iron flash making
	For heat treating metals
	In welding processes
	As a reducing agent
Flat Glass Production	To prevent oxidation
Electronics Manufacturing	As a reducing agent
	For etching
Medical	In the production of hydrogen peroxide
	As an antioxidant
	In cooling systems for MRI machines

Fig.1.5.3: Uses of hydrogen in other industries

- Tell them that, in transport industry, green hydrogen can be used in:

Space Exploration	Aviation
Logistics	Public Transportation

Fig.1.5.4: Uses of hydrogen in transport industry

- Tell them that, in power sector, green hydrogen can be used as follows:

As a cooling agent in thermal power generators	To generate electricity

Fig.1.5.5: Uses of hydrogen in power sector

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank papers and pens to each group.
- Ask the groups to make five columns on the paper and list the uses of hydrogen in various industries.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 1.6: Technology Options for Production of Green Hydrogen

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the electrolysis process
2. Identify the different types of electrolyzers

Ask

Ask the participants if they know they about the electrolysis process.

Demonstrate

The electrolysis process

Notes for Facilitation

- Explain the electrolysis process to the participants.
- Tell them that there are four types of electrolysis process:

Alkaline Electrolysis (AEL)	Proton Exchange Membrane (PEM) Electrolysis
Solid Oxide Electrolysis (SOE)	Anion Exchange Membrane Electrolysis (AEM)

Fig.1.6.1: Types of electrolysis

- Tell them about each electrolysis process in detail.

UNIT 1.7: Renewable Energy Sources for Large Scale Production of Green Hydrogen through water electrolysis

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the concept of renewable energy.
2. Explain various sources of renewable energy

Ask

Ask the participants if they know they can tell some sources of renewable energy.

Notes for Facilitation

- Introduce the topic of renewable energy sources.
- Tell them that the main sources of renewable energy are as follows:

Solar	Hydro	wind
Tidal	Geothermal	Biomass

Fig.1.7.1: Sources of renewable energy

- Explain the principle and the working of each type of renewable energy source in detail.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank papers and pens to each group.
- Ask the groups to make six columns on the paper and list the working of each type of renewable energy source.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

Ask

Ask the participants if they know they can tell some benefits of renewable energy.

Notes for Facilitation

- Tell them that the benefits of using renewable energy sources over traditional fossil fuel are as follows:

Reduced Greenhouse Gas Emissions	Inexhaustible Resources	Reduces dependency on imported fuels
Rural Development	Reduced Water Usage	Community Empowerment
Long-Term Sustainability	Low Operating and Maintenance Costs	Job Opportunities

Fig.1.7.1: Benefits of renewable energy

- Explain about of each benefit of renewable energy source in detail.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to make six columns on paper and list the benefits of renewable energy sources.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 1.8: Emerging Entrepreneurial Opportunities Across Green Hydrogen Value chain

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss in brief the emerging entrepreneurial opportunities across Green Hydrogen Value chain
2. Identify opportunities to various renewable energy sources.

Ask

Ask the participants if they know the term – entrepreneurial opportunities across Green Hydrogen Value chain

Notes for Facilitation

- Tell them about each stage in the green hydrogen value chain and where entrepreneurial opportunities lie.

Key Topics:

- **Hydrogen Production**: Electrolysis technology, feedstock, and renewable energy integration.
- **Hydrogen Storage**: Compressed, liquefied, and metal hydride storage solutions.
- **Hydrogen Distribution**: Pipeline, tankers, refueling stations, and grid integration.
- **Hydrogen Utilization**: Applications in industry, transport, power generation, and emerging sectors.
- **End-Use Markets**: Commercial and industrial applications (energy, automotive, refining, etc.).

Notes:

- Explain how each part of the value chain is interlinked and represents unique entrepreneurial opportunities.
- Provide real-world examples and case studies from startups or established players.

Notes for Facilitation

Renewable Energy	Opportunities
Solar Energy	India has abundant sunlight throughout the year, making it a great place for solar power generation. Opportunities include solar panels on rooftops, solar farms in open areas, and solar-powered appliances.
Wind Energy	Coastal regions and certain inland areas in India have strong wind resources. Opportunities involve setting up wind farms and harnessing wind energy for electricity generation.
Hydropower	India has rivers and hilly terrain suitable for hydropower projects. Opportunities include building small and large-scale hydropower plants to generate electricity.
Biomass Energy	India has a vast agricultural sector, which can provide biomass sources like crop residues and organic waste. Opportunities involve converting biomass into biofuels and using it for power generation.
Geothermal Energy	Certain regions in India have geothermal potential. Opportunities include exploring and utilizing geothermal energy for heating and electricity.
Tidal and Wave Energy	India has a long coastline with potential for tidal and wave energy generation. Opportunities involve deploying technologies to harness energy from tides and waves.
Waste-to-Energy	India's urban areas produce significant amounts of waste that can be converted into energy through waste-to-energy plants. Opportunities include waste management and energy generation.
Microgrids	Setting up microgrids in remote areas can provide clean and reliable energy access. Opportunities involve decentralized renewable energy solutions.
Energy Efficiency	Opportunities also lie in improving energy efficiency across various sectors, reducing energy wastage, and promoting energy-efficient technologies.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to list the key highlights of entrepreneurial opportunities across Green Hydrogen Value chain
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 1.9: Role and Responsibilities of a Green Hydrogen Plant Entrepreneur

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the role and responsibilities of a Green Hydrogen Plant Entrepreneur.

Ask

Ask the participants if they know the role and responsibilities of green hydrogen plant technician.

Notes for Facilitation

- Tell them about the career progression for Green Hydrogen Plant Entrepreneur.
- Tell them that a Green Hydrogen Plant Entrepreneur has a very important role to play in the green hydrogen generation facilities.
- Tell them in detail about their responsibilities, which are as follows:

Market Analysis
Client Needs Assessment
Client Relationship Management
Feasibility Assessment
Proposal Development
Project Planning
Component Selection and Installation
Regulatory Compliance
Risk Management
Resource Management
Monitoring and Reporting
Environmental Sustainability

Fig.1.9.1: Role and Responsibilities of a Green Hydrogen Plant Entrepreneur

- Tell them that, in addition to technical skills, a green plant technician should possess the following attributes:

Sincerity	Diligence	Hardworking
Attentiveness	Quality Focus	Safety Consciousness
Adaptability	Team Player	Problem Solving

Fig.1.9.2: Attributes of a Green Hydrogen Plant Entrepreneur

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to list the responsibilities of a green plant entrepreneur.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

2. Components of Green Hydrogen Plant and its Layout

Unit 2.1 - Key Components of the Green Hydrogen Plant

Unit 2.2 - Functions of Key Components and Their Fundamental Principles

Unit 2.3 - Types of Electrolyzers - Brief Description

Unit 2.4 - Overall Layout of the Plant and Sparking Items

Unit 2.5 - Read, Interpret and Understand the Relevant Drawings and O&M Manuals of the Plant Equipment

Unit 2.6 - Key Material and Safety Codes, Technology Protocols and Standards Applicable in Green Industry

Unit 2.7– Key Technical Insights for Setting Up and Running Successful Enterprise.

SGJ/N4101

Key Learning Outcomes

At the end of this module, you will be able to:

1. Identify all key components of the Green Hydrogen plant including electrical, mechanical and civil components
2. Discuss functions of the key components of electrolyzer stack, renewable power plant, feed water supply unit, gas separator, transformer and rectifier, gas compression unit, etc.
3. Explain overall layout of the plant
4. Read and interpret various electrical codes, data sheet, safety features etc relevant to the plant.
5. Discuss working principles of main components including electrolyzer stack, gas separator, power source, etc.,
6. Discuss key technical insights of Green hydrogen system for setting up and running successful enterprises
7. Discuss how to perform SWOT analysis
8. Discuss safety aspects to be considered while preparing the layout of hydrogen production plant for prevention from fire caused by sparking, e.g. to ensure safe distance of hydrogen from electrical devices like MCBs, switches.

UNIT 2.1: Key Components of the Green Hydrogen Plant

Unit Objectives

At the end of this unit, you will be able to:

1. Identify key components of green hydrogen plant
2. Categorize electrical, mechanical and civil components

Ask

Ask the participants if they know what a green hydrogen plant is.

Notes for Facilitation

- Tell them that a green hydrogen production plant is a facility that generates hydrogen gas using renewable energy sources, primarily through the process of water electrolysis.
- Tell them that the key components of a green hydrogen plant are as follows:

Electrical
- Transformer
- Rectifier
- Electrolyzer
- Electrolytre

Mechanical
- Demineralization (DM) water plant
- Piping, pumps, valves
- Compressors
- Hydrogen separator
- Oxygen separator

Civil
- Plant buildings, Administrative building
- Foundation of equipment, panels, feed water supply
- Earthing pits, trenches etc. in the plant including substation

Fig.2.1.1: Components of a green hydrogen plant

- Tell them in detail about each type of components.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to make three columns, titled, Electrical, Mechanical and Civil
- Ask them to list the key components for each of the three.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 2.2: Functions of Key Components and Their Fundamental Principles

Unit Objectives

At the end of this unit, you will be able to:

1. Identify key components and state their primary functions.
2. Explain the fundamental principles governing each component.
3. Demonstrate how components operate within the overall system.
4. Analyze interactions and performance dependencies among components.
5. Assess efficiency and operational limitations of components.
6. Propose improvements based on functional principles.

Ask

Ask the participants if they name some electrical components used in green hydrogen plants.

Notes for Facilitation

- Tell them that following electrical components are used in green hydrogen plants:

Transformer	Rectifier	Electrolyzer Stack

Fig.2.2.1: Electrical components used in green hydrogen plants

- Tell them in detail about the function of each component.
- Tell them about the various mechanical components used in green hydrogen plants, such as:

Compressors	Pumps	Valves	Ventilation
Piping	Firefighting system	DM water plant	Hydrogen separator

Fig.2.2.2: Mechanical components used in green hydrogen plants

- Tell them in detail about the function of each component.
- Solution to activity: Match the components of an electrolyzer with their functions
    - A-G, B-H, C-F, D-E

Ask

Ask the participants if they know the fundamental principles of key components in a green hydrogen plant.

Notes for Facilitation

- Tell them that every electrical component plays a crucial role in the operation of a green hydrogen plant.
- Tell them that an electrolyte contains the following components:

Electrodes	Electrolyte	Membrane	Catalysts

Fig.2.2.3: Components of electrolyzer

- Tell them about the function of each component.
- Tell them that the inputs to an electrolyzer are:
    a. Demineralized water
    b. Electrical energy
- Tell them about the basic chemical equation that depicts the electrolysis process.
- Explain to them the parts of:
    a. Alkaline electrolyzer
    b. PEM electrolyzer
- Tell them about electrolyzer stacks.
- Tell them that the electrolyzer stack works by application of an electric current to split water molecules into hydrogen and oxygen gases through an electrochemical reaction.
- Ask them if they know that pure water cannot conduct electricity.
- Tell them that to make water conduct electricity, an alkali or an acid is added. This solution is known as an electrolyte.
- Tell them about the electrolytes used in respective electrolyzers.
- Tell them that the electricity in an electrolyzer is conducted with the help of two electrodes—an anode and a cathode.
- Explain that a catalyst is a substance that accelerates a chemical reaction.
- Tell them that a membrane is a thin separator, used in electrolyzers, to separate anode and cathode loops.
- Explain that a diaphragm is a flexible and relatively thick barrier that separates different compartments within the electrolyzer.
- Explain that the bipolar plates provide the electrical connections between one cell and the next cell of the stack.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to make four columns, titled, Electrodes, Electrolyte, Membrane and Catalyst

- Ask them to list the functions of each of them.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

Ask

Ask the participants if they know what demineralization of water is.

Notes for Facilitation

- Introduce the concept of Demineralized (DM) water.
- Tell them that demineralization (DM) is the process of removing all the soluble minerals from water.
- Tell them that the following electrical components are used in DM water plants:

Sand Filter	Cationic bed	Anionic bed	Mixed bed

Fig.2.2.4: Components used in DM water *plants*

- Tell them about the function of each component.
- Tell them about the function of a hydrogen gas collector drum.
- Tell them about hydrogen separators and oxygen separators.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to list the components of DM water plant and the function of each component.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 2.3: Types of Electrolyzers: Brief Description

Unit Objectives

At the end of this unit, you will be able to:

1. Describe different types of electrolyzers

Ask

Ask the participants if they know the types of electrolyzers.

Notes for Facilitation

- Tell them that there are four main types of electrolyzers; each has its own advantages and disadvantages.
- Tell them that which electrolyzer is suitable for a particular application depends on factors such as:

Efficiency	Operating Conditions	Scalability	Cost

Fig.2.3.1: Factors that distinguish electrolyzers

- Tell them that the main types of electrolyzers are:

Polymer Electrolyte Membrane (PEM) Electrolyzers	Alkaline Electrolyzers
Solid Oxide Electrolyzers	Anion Exchange Electrolyzer

Fig.2.3.2: Factors that distinguish electrolyzers

Notes for Facilitation

- Tell them that in a Proton Exchange Membrane (PEM) electrolyzer, a solid polymer membrane serves as the electrolyte.
- Explain in detail the working of PEM electrolyzer.
- Tell them that Alkaline electrolyzers use a liquid alkaline solution of potassium or sodium hydroxide as the electrolyte.
- Tell them about the advantages and disadvantages of an alkaline electrolyzer.
- Tell them that in Solid oxide electrolyzers, a solid ceramic material is used as the electrolyte.
- Explain in detail the working of Solid oxide electrolyzers.
- Tell them that anion exchange membrane electrolyzer uses solid alkaline membrane as an electrolyte.
- Explain in detail the working of anion exchange membrane electrolyzer.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to make four columns.
- Ask them to list the different types of electrolyzers and also the basic features of each of them.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 2.4: Overall Layout of the Plant and Sparking items

Unit Objectives

At the end of this unit, you will be able to:

1. Explain overall layout of the plant
2. Identify the key components in the layout
3. Discuss Sparking items e.g. MCBs, switches to be installed isolated from the hydrogen production area

Notes for Facilitation

- Do a quick recap of the layout of green hydrogen plant, as discussed in unit 2.1.
- Tell them about the optimum utilization of the resources in various processes.
- Tell them about sparking items.
- Explain to them that the rate at which hydrogen is produced in an electrolyzer is directly proportional to the amount of electric charge that is engaged in the electrolysis process.
- Tell them about the importance of installing switchgears to ensure safety.

UNIT 2.5: Read, Interpret and Understand the Relevant Drawings and O&M Manuals of the Plant Equipment

Unit Objectives

At the end of this unit, you will be able to:

1. Identify different types of technical drawings and O&M manuals related to plant equipment.
2. Explain symbols, notations, and standard conventions used in engineering drawings.
3. Interpret layout, P&ID, electrical, and mechanical drawings accurately.
4. Analyze equipment specifications and operational procedures in O&M manuals.
5. Evaluate compliance with safety and operational guidelines.
6. Apply information from drawings and manuals to support safe and efficient plant operation.

Ask

Ask the participants if they know about engineering drawings.

Demonstrate

Show a sample of Engineering drawing.

Notes for Facilitation

- Tell them that an engineering drawing is a type of graphical representation of design concepts and technical specifications and communicates ideas and information.
- Tell them that an engineering drawing contains one or more of the following types of information:

Dimensions	Geometry	Tolerances
Material type	finish	Hardware

Fig.2.5.1: Information in engineering drawing

- Tell them that the lower right-hand corner of the drawing contains the information blocks.
- Tell them that the main information blocks are:
  - Title block
  - Revision block
- Tell them about the Bill of Materials (BOM).
- Tell them about the three types of lines:
  - Visible lines
  - Hidden lines
  - Centre lines
- Show them how to read an engineering drawing.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give an engineering drawing to each group.
- Ask them to list the different information contained in the drawing.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

Ask

Ask the participants if they know about O&M manuals.

Demonstrate

Show a sample O&M manual.

Notes for Facilitation

- Tell them that an Operation and Maintenance (O&M) manual is a document that provides essential details about the upkeep of equipment and machinery.
- Tell them about the general information that is included in O & M manual.
- Tell them about the codes and standards that they will need to refer manuals in case of confusion or doubt in day-to-day operations, routine maintenance etc.
- Tell them about The Indian Standard 15201: 2002, 'Hydrogen – Code of Safety'.

UNIT 2.6: Key Material and Safety Codes, Technology Protocols and Standards Applicable in Green Industry

Unit Objectives

At the end of this unit, you will be able to:

- Explain key material and safety codes, Technology protocols and standards as applicable in Green hydrogen industry

Ask

Ask the participants if they know the hazards associated with hydrogen and the preventive measures for controlling the hazards.

Notes for Facilitation

- Start the session be telling them about the key materials used in different components of electrolyzers.
- Introduce the topic of safety hazards.
- Tell them that the main safety challenges involved in working with hydrogen is fire and explosion hazard.
- Tell them that the risk of fire depends on the following factors:

Hydrogen leakage	Flammability Range	Low Ignition Energy

Fig.2.6.1: Factors responsible for fire risk

- Explain each factor in detail.
- Explain to them about the Indian Standards on Hydrogen.

Ask

Ask the participants if they know about the personal protective equipment (PPE).

Demonstrate

- Demonstrate the correct way of wearing PPE.

Notes for Facilitation

- Introduce the topic of personal protective equipment (PPE).
- Explain that it is very important for them to use safety equipment while working as they provide protection to various parts of their body.
- Explain the full form of the term 'PPE'.
- Tell them that:
    - Goggles provide them with eye protection from hazards such as electric sparks, minute flying particles, dust and so on.
    - Helmets provide protection from injury to head from falling objects and slipping and tripping.
    - Gloves provide hand protection from hazards such as harmful substances, cuts and abrasions, chemical or electrical burns and so on.
    - Shoes provide foot protection from hazards such as falling heavy objects, hot, wet and slippery floor and so on.
    - Ear plugs provide hearing protection from extreme noise levels.

Team Activity

- Divide the class into pairs of two participants each.
- Give them a PPE kit.
- Ask them to demonstrate the correct way of wearing PPE.

Ask

Ask the participants if they know the safety guidelines to be followed while storing, handling and labelling hydrogen.

Notes for Facilitation

- Explain that it is very important for them to follow the safety guidelines while storing, handling and labelling hydrogen.
- Tell them about:
  - Storage Guidelines
  - Handling Tools
  - Safety Measures in Storage Areas
    - Ventilation
    - Indoor Storage Precautions
    - Outdoor Storage Precautions
    - Bulk Storage (Non-refrigerated)
- Tell them about the checks to be conducted before filling hydrogen cylinders.
- Tel them about the importance of labelling the hydrogen cylinders properly.
- Tell them that they should do a regular inspection of hydrogen containers to detect any potential leaks.
- Tell them about the precautions that should be followed while handling hydrogen leakage.
- Tell them that the waste generated during the hydrogen plant operations should be properly disposed of to prevent any potential hazard.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give one of the following topic to each group.
- Ask them to list the guidelines to be followed.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

- Topic 1: Guidelines while storing, handling and labelling hydrogen.
- Topic 2: Guidelines while handling Hydrogen cylinders.
- Topic 3: Guidelines for handling spillage or leakage of Hydrogen.

Ask

Ask the participants if they can list some energy sources that can cause ignition to hydrogen.

Demonstrate

The firefighting equipment

Notes for Facilitation

- Introduce the topic of fire prevention and fire fighting.
- List the various energy sources that can cause ignition to hydrogen.
- Tell them about the precautions that must be taken to prevent potential fire hazard.
- Tell them that there are two types of hydrogen spills:

Unignited Spills	Ignited Spills

Fig.2.6.2: Types of hydrogen spills

- Tell them about the ways to handle each type of spill.
- List the best practices to deal with the hydrogen fire.
- Tell them that there are different types of fire extinguishers.
- Tell them about:
    - Classification of fire
    - Symbols denoting the fire extinguishers
    - Extinguishing agents used in each type

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank papers and pens to each group.
- Ask the groups to make five columns on the paper and list the different types of fire extinguishers based on the classification of fire, their symbols, and extinguishing agents used in extinguishing the fire.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

Ask

Ask the participants if they know how to provide first aid if someone inhales excessive hydrogen.

Demonstrate

First aid that to be provided if someone inhales excessive hydrogen.

Notes for Facilitation

- Introduce the topic of training and health monitoring.
- Tell them about the importance of undergoing training relevant to their roles.
- List the topics that must be covered in the training.
- Tell them that in order to identify any potential exposure risk, they must monitor their health and hygiene. And undergo periodic physical examination.
- Tell them about the first aid that should be provided if someone inhales excessive hydrogen.
- After the demonstration, ask the participants if they have any queries.
- Tell them about the steps to be taken if someone comes in direct contact with liquid hydrogen.

Team Activity

- Divide the class into pairs.
- Ask one of the volunteers to be the technician and other to be his coworker who has inhaled hydrogen.
- Ask the pairs to demonstrate how to provide first aid.
- Give the groups 5 minutes for the activity.

Ask

Ask the participants if they know about emergency action plan that should be followed during any crisis.

Demonstrate

A sample of emergency action plan.

Notes for Facilitation

- Introduce the topic of emergency action plan.
- Tell them that an emergency action plan is a set of guidelines outlining procedures to follow during crises, ensuring safety and effective response in various emergency situations.
- Show the guidelines for the emergency action plan.
- Tell them about the safety instructions for hazardous area.

UNIT 2.7: Key technical insights for setting up and running Successful Enterprise

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss key technical insights of Green hydrogen system for setting up and running successful enterprises
2. Discuss how to perform SWOT analysis

Ask

Ask the participants if they know the technical insights of setting up and running successful enterprises.

Demonstrate

Perform SWOT analysis for Green Hydrogen sector.

Notes for Facilitation

Introduce the key technical insights of setting up and running successful enterprises in India.

- Renewable Energy Integration: India boasts abundant solar and wind resources. Leveraging these renewables for electrolysis, the heart of Green Hydrogen production, is cost-effective and sustainable. Understanding how to efficiently integrate renewable energy sources into the hydrogen production process is essential.
- Electrolysis Technology: India's climate and energy infrastructure can influence the choice of electrolysis technology. Proton Exchange Membrane (PEM) electrolyzers may be preferable for regions with variable renewable energy, while Alkaline electrolyzers can be cost-effective in areas with stable energy supply.
- Hydrogen Storage: Efficient hydrogen storage is key, especially for remote regions with intermittent energy supply. Exploring high-pressure tanks, liquid organic hydrogen carriers (LOHCs), or metal hydrides can address India's diverse storage needs.
- Safety Protocols: India's regulatory environment demands rigorous safety standards. Understanding and implementing comprehensive safety protocols is vital due to hydrogen's flammability.
- Regulatory Compliance: Navigating India's evolving regulatory landscape, which promotes clean energy, requires a clear understanding of compliance requirements. Staying abreast of incentives, permits, and environmental standards is essential.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the customer and others would ask questions.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

3. Key technical and entrepreneurial aspects for supporting growth and business development green hydrogen production

SGJ/N1818

Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss process for setting up hydrogen production plant including required registration, availing low-cost financing from Financial Institutions/Venture Capital/Banks, government subsidies, green power, freewheeling/banking of renewable energy etc.
2. Discuss various evolving opportunities for setting up of hydrogen production plants at different sites, technologies, services, finance and market ecosystem
3. Discuss how to identify the key ingredients of business plan for land, water, renewable energy, finance, technology and market mechanism
4. Discuss how to assess the potential and demand of Hydrogen in different sectors i.e. transport, power, building, industry, etc.
5. Discuss all key aspects of a Detailed Project Report (DPR) and explain how to develop a bankable DPR which includes market demand, storage, technology and finance for hydrogen production
6. Discuss various parameters for calculating lifecycle cost of hydrogen production, storage system, along with cash flow, annual savings, payback period, IRR, inflation, increase in fossil fuel cost, sensitivity analysis by varying costs of renewable energy, water, man power, and such other factors etc. and how to include those in the detailed project report (DPR)
7. Discuss key aspects to successfully install, operate, safely store and market Hydrogen as Energy source.
8. Discuss O&M related cost for the system. If required, possibility of Annual Maintenance Contract (AMC) of electrolyser from supplier/ manufacturers, and other systems may be explored.
9. Explain different taxation and related compliances to setting up and operations
10. Discuss National Green Hydrogen policy and corresponding incentives/subsidies available and procedures for availing the same.
11. Explain how evolving start-ups and new ventures boost-up hydrogen economy.

UNIT 3.1: Setting Up Green Hydrogen Production Plant

Unit Objectives

At the end of this unit, you will be able to:

1. Explain regulatory and approval requirements for hydrogen plant setup.
2. Identify financing sources, subsidies, and renewable energy mechanisms.
3. Recognize emerging opportunities in hydrogen technologies and markets.
4. Identify key components of a hydrogen business plan.
5. Assess hydrogen demand across major sectors.
6. Outline essential elements of a bankable Detailed Project Report (DPR).

Ask

Ask the participants if they have any idea for setting up Hydrogen Production Plant.

Explain

Explain the basics steps involved in setting up a hydrogen production plant in India.

Tell the learner about the Registrations Required for Setting Up Hydrogen Production Plant.

Tell the initiatives and schemes related to green hydrogen production in India.

Tell the learner about the evolving opportunities in Hydrogen Production in India.

Explain Key Ingredients of a Business Plan. Explain key factors play an essential role in deciding whether a project will turn into a success. Explain information to be included in a DPR.

Explain advantages of DPR.

Demonstrate

Create a bankable DPR. To know more about Detailed Project Report, watch the video from the link given below.
https://www.youtube.com/watch?v=Ke_jwp_lskg

Notes for Facilitation

1. Introduce the topic. Tell the learners about the Key Ingredients of a Business Plan.
   A well-structured business plan typically includes the following key ingredients:

   - **Executive Summary**
   - **Company Description**
   - **Market Research and Analysis**
   - **Products or Services**
   - **Marketing and Sales Strategy**
   - **Operations Plan**
   - **Financial Plan**
   - **Management and Team**
   - **Risk Assessment**
   - **Sustainability and Environmental Impact**

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank papers and pens to each group.
- Give them a question asking them to prepare DPR.
- Ask the group to discuss and write down the answers.
- Give the groups 15 minutes for the activity.

UNIT 3.2: Cost Estimation

Unit Objectives

At the end of this unit, you will be able to:

- Discuss various parameters for calculating lifecycle cost of hydrogen production, storage system, along with cash flow, annual savings, payback period, IRR, inflation, increase in fossil fuel cost, sensitivity analysis by varying costs of renewable energy, water, man power, and such other factors etc. and how to include those in the detailed project report (DPR)
- Discuss key aspects to successfully install, operate, safely store and market Hydrogen as Energy source.
- Discuss O&M related cost for the system. If required, possibility of Annual Maintenance Contract (AMC) of electrolyser from supplier/ manufacturers, and other systems may be explored.

Explain

Explain the formula used to calculate the lifecycle cost (LCC) of a green hydrogen plant.

Tell the learner about the Key Aspects to Successfully Install, Operate, Safely Store and Market Hydrogen as Energy Source.

Tell the learner about operation and maintenance (O&M) costs for a hydrogen production system. Tell the learner about Annual Maintenance Contract (AMC).

Notes for Facilitation

Tell the learner about formula used to calculate the lifecycle cost (LCC) of a green hydrogen plant involves summing all relevant costs and benefits over the project's entire lifespan. The general formula for LCC can be expressed as:

- LCC = Initial Investment + Operating and Maintenance Costs - Annual Savings + Salvage Value

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 3.3: Taxation and Related Compliances

Unit Objectives

At the end of this unit, you will be able to:

1. Explain different taxation and related compliances to setting up and operations

Explain

Tell the learner about the Taxation and Related Compliances in India.

Explain the various compliances relevant to green hydrogen projects.

Tell the learner about Goods and Services Tax (GST).

Notes for Facilitation

Tell the learner about various taxation relevant to green hydrogen projects:

- Goods and Services Tax (GST):

o GST is applicable to the supply of goods and services in India, including equipment, materials, and services related to the construction and operation of a green hydrogen plant.

o Businesses need to register for GST, file periodic GST returns, and pay applicable GST on their transactions.

- Income Tax:

o Companies operating green hydrogen plants are subject to corporate income tax in India. The current corporate tax rate varies depending on the type of business entity and annual turnover.

o Businesses need to file annual income tax returns and comply with transfer pricing regulations for international transactions.

- Customs Duties and Import Taxes:

o Importing equipment, machinery, and components for the green hydrogen plant may incur customs duties and import taxes.

o Businesses should understand the customs duties applicable to specific imports and comply with customs regulations.

- State-Specific Taxes:

o Different states in India may have their own state-level taxes, such as state GST (SGST) and state-specific incentives or concessions.

o Businesses should be aware of state-level tax implications and incentives when selecting the plant's location.

UNIT 3.4: Policies and Corresponding Incentives/Subsidies

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss National Green Hydrogen policy and corresponding incentives/subsidies available and procedures for availing the same
2. Explain how evolving start-ups and new ventures boost-up hydrogen economy.

Explain

Tell the learner about the aim of a National Green Hydrogen Policy.

Explain the various subsidies available for Renewable energy Projects.

Tell the learner about role of new ventures and start-ups in the green hydrogen economy.

Notes for Facilitation

Tell the learners about following schemes

• Government Grants: Some government programs and schemes may offer grants to support the development and deployment of green hydrogen technologies.

o Procedure: To access government grants, project developers typically need to apply through a competitive application process, which may involve submitting project proposals, budgets, and compliance with specific grant requirements.

• Renewable Energy Certificates (RECs): Green hydrogen production using renewable energy sources may qualify for RECs, which can be traded and used to demonstrate the use of renewable energy.

o Procedure: Project developers can participate in REC trading platforms and follow the standard procedures for trading and certification.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

4. Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production

SGJ/N1820

Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss the Electrolyser Type (PEM, AE, SOEC)
2. Explain the technical specification of Electrolyser (PEM, AE and SOEC)
3. Explain the Major components of Electrolyser
4. Discuss the Operation of PEM and AE and SOEC electrolyser.
5. Discuss Comparison between above electrolyzers with reference to the life, cost, efficiency, electricity consumption etc.
6. Discuss the Tools required for Installation, dismantling, removal of components of PEM and AE /Alkaline water electrolysers.
7. Explain the inputs / outputs of Electrolyzer
8. Discuss in detail the input renewable power from various sources and its integration with electrolyser.
9. Discuss how to oversee the Installation of Parts and Components of Electrolyser
10. Discuss step by step process for Installation of Electrolyser
11. Explain the Challenges associated with Hydrogen in storage, handling and transportation
12. Discuss how to select and install hydrogen compression and storage system
13. Discuss the precautions required to compress and store hydrogen.
14. Discuss the safety guidelines to be followed as per industry standard

UNIT 4.1: Types of Electrolyzers, their technology maturity and Technical Specifications

Unit Objectives

At the end of this unit, you will be able to:

- Discuss the Electrolyser Type (PEM, AE, SOEC)
- Explain the technical specification of Electrolyser (PEM, AE and SOEC)
- Explain the Major components of Electrolyser
- Discuss the Operation of PEM and AE and SOEC electrolyser.
- Discuss Comparison between above electrolyzers with reference to the life, cost, efficiency, electricity consumption etc.

Ask

Ask the participants if they know the meaning of the term 'electrolyzers'.

Notes for Facilitation

- Tell the participants about electrolyzers.
- Tell them about the technological status of electrolyzers.
- Tell the participants about the several types of electrolyzers.

Say

- Explain to them about the technological status of electrolyzers saying that it is rapidly evolving and new developments are being made all the time.
- Tell them about the cost of electrolyzers is expected to decrease significantly in the coming years as demand for hydrogen is increasing and technological improvements are made. Electrolyzers are a key technology in the development of a clean energy economy, and they can be used to produce hydrogen from renewable energy sources.
- Electrolyzers are classified into four main types:

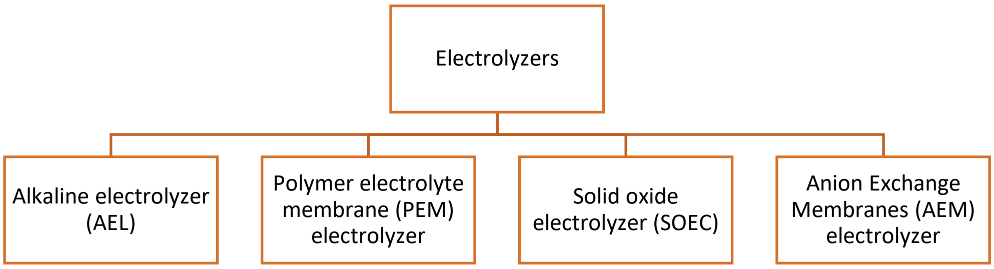


Fig 4.1.1: Types of electrolyzers

Ask

- Ask the participants about types of electrolyzer.

Notes for Facilitation

- Explain about the types of electrolyzers one by one.
- Start with the alkaline electrolyzer (AEL).
- Explain Polymer Electrolyte membrane (PEM) electrolyzer.
- Tell them about the Solid Oxide Electrolyzer (SOEC).
- Tell them about the Anion Exchange Membrane (AEM).

Explain

- Begin by describing how an alkaline electrolyzer (AEL) works by using an alkaline solution as the electrolyte, such as potassium hydroxide (KOH) or sodium hydroxide (NaOH). Anode and cathode, the two electrodes that make up a cell, are often separated by a membrane or diaphragm. Water molecules split into hydrogen and oxygen atoms at the electrodes when electricity is applied to the electrolyzer.
- At the cathode, hydrogen atoms are gathered, while at the anode, oxygen atoms are gathered. The hydrogen and oxygen gases are kept apart by a membrane or diaphragm, which prevents them from recombining.
- They require more electricity to produce the same amount of hydrogen.
- AELs are a crucial piece of technology for the growth of a clean energy economy. Through the utilization of renewable energy sources like solar and wind energy, hydrogen may be produced.

Demonstrate

- Show them an alkaline electrolyzer.
- Show the working of alkaline electrolyzer.

Explain

- Begin by describing how the Polymer Electrolyte Membrane (PEM) Electrolyzer conducts protons from the anode to the cathode while electrically insulating the electrodes.
- They are less effective than PEM electrolyzers, which can work at greater temperatures.
- They can thus be used for a greater variety of purposes, such as the production of hydrogen for fuel cells. AELs are anticipated to become much less expensive as hydrogen demand rises.

Components of PEM electrolyzer

- Anode: such as iridium.
- Cathode: such as platinum
- Electrolyte: made of Nafion
- Bipolar plates:made of titanium or graphite

Fig 4.1.2: Components of PEM electrolyzer

Explain

- Start by explaining that a Solid Oxide Electrolyzer (SOEC) uses a solid oxide electrolyte to conduct oxygen ions from the cathode to the anode while insulating the electrodes electrically. It operates at very high temperatures (500 and 800 °C), makes SOECs very efficient, but it also requires more expensive materials and construction.
- SOECs typically consist of the following components:

- Anode: Such as nickel-yttria-stabilized zirconia (YSZ).
- Cathode: Such as lanthanum strontium manganite (LSM).
- Electrolyte: Such as YSZ, that conducts oxygen ions from the cathode to the anode.
- Bipolar plates: Typically made of stainless steel or nickel-chromium alloy.

Fig 4.1.3: Components of SOECs

Explain

- Start by explaining that an **Anion Exchange Membrane (AEM)** electrolyzer uses an anion exchange membrane to conduct hydroxide ions from the cathode to the anode while insulating the electrodes electrically.
- AEM electrolyzers are like PEM electrolyzers, but they have a number of advantages, such as:
  - They can use non-precious metal catalysts, which are less expensive and more abundant.
  - They are less sensitive to impurities in the water feed.
  - They are more durable than PEM electrolyzers.

Components of Anion Exchange Membrane (AEM) electrolyzer

Anode: Such as nickel.	Cathode: Such as nickel.	Electrolyte: such as poly (phenylene sulfide) (PPS).	Bipolar plates

Fig 4.1.4: Components of AEM electrolyzer

Explain

- Explain to them the key Technical Specifications of various Types of Electrolyzer
- The Four main electrolyzers; PEM, AE AEM, SOEC are at different stages of their development and commercialization.
- List the points of comparisons between Alkaline: MW and PEM electrolyzers

Activity

- Outline differences in PEM, AE, AEM and SOEC Electrolyzer and illustrate their schematics.

Explain

- Tell them about the Operating principles of PEM, AE and SOEC electrolyzer.

Proton Exchange Membrane (PEM) Electrolyzer:

- Requires pure water and operates at low temperatures.
- Fast response, suitable for intermittent power sources.
- High efficiency but sensitive to impurities.

Alkaline Electrolyzer (AE):

- Requires a conductive separator between electrodes.
- Hydroxide ions migrate, forming hydrogen at the cathode and oxygen at the anode.
- Robust but limited flexibility in operation.

Solid Oxide Electrolysis Cell (SOEC):

- Employs solid oxide electrolyte that conducts oxygen ions.
- Oxygen ions migrate from the cathode to anode, where they react with hydrogen to form water.
- Requires external heat, usually from high-temperature sources.

Fig 4.1.5: Operating principle

- The following table shows some specific technical specifications for different types of electrolyzer cells:

Type of electrolyzer	Cell voltage (V)	Current density (A/cm2)	Cell efficiency (%)	Operating temperature (°C)	Operating pressure (ATM)
Alkaline electrolyzer	1.5-1.8	100-500	60-70	25-80	1-10
Proton exchange membrane electrolyzer	1.6-1.9	100-1000	70-80	25-100	1-100
Solid oxide electrolyzer	0.8-1.2	100-1000	70-80	500-800	1-100

Table 4.1.1: Some specific technical specifications

UNIT 4.2: Tools and Equipment for Installation

Unit Objectives

At the end of this unit, you will be able to:

1. Identify suitable tools and equipment required for Installation of electrolyzer, Plant & Machineries conforming to relevant technical sheets, safety and technical standards for proper execution of work.
2. Understand safety guidelines and applicable standards for proper execution of work.

Ask

- Ask the participant if they ever have installed or seen the installation of an electrolyzer.

Notes for Facilitation

- Tell them the equipment required for the installation.
- Also tell them to take required safety measures during the installation.

Explain

- Start by explaining that the following tools and equipment are needed for the installation of an electrolyzer:

Safety gear: Safety glasses, gloves, boots, and hard hat

Lifting equipment: Hoist or crane to lift the electrolyzer into place

Tools: Wrenches, screwdrivers, and other tools to connect the electrolyzer to the water, electricity, and hydrogen lines.

Materials: Piping, valves, fittings, and other materials to connect the electrolyzer to the water, electricity, and hydrogen lines

Test equipment: Pressure gauges, flow meters, and other equipment to test the electrolyzer for leaks and proper operation

Fig 4.2.1: Tools and equipment which are required for installation

- In addition to the above, the following tools and equipment may also be needed, depending on the specific type of electrolyzer being installed:

Alkaline electrolyzer:	Anode and cathode baskets, diaphragm or membrane, and electrolyte solution
Polymer electrolyte membrane (PEM) electrolyzer:	Anode and cathode gas diffusion electrodes, polymer electrolyte membrane, and bipolar plates
Anion exchange membrane (AEM) electrolyzer:	Anode and cathode gas diffusion electrodes, anion exchange membrane, and bipolar plates.
Solid oxide electrolyzer (SOEC):	Anode and cathode porous electrodes, solid oxide electrolyte, and bipolar plates

Fig 4.2.2: Tools and equipment may also be needed, depending on the specific type of electrolyzer being installed

NOTE: It is important to consult the manufacturer's instructions for the specific electrolyzer being installed to ensure that all of the necessary tools and equipment are available, and that the installation is performed correctly.

Demonstrate

- Show the participant how to install an electrolyzer.

Notes for Facilitation

➢ Tell them about some tips for installation:

- Make sure that the installation site is well-ventilated and that there is adequate space around the electrolyzer for maintenance and operation.
- Follow the manufacturer's instructions carefully when installing the electrolyzer.
- Use the correct piping, valves, and fittings for the specific type of electrolyzer being installed.
- Test the electrolyzer for leaks and proper operation before putting it into service.

Electrolyzers are complex devices, so it is important to have a qualified technician.

UNIT 4.3: Inputs/Outputs of an Electrolyzer System and Calculate Losses and Equipment Efficiency

Unit Objectives

At the end of this unit, you will be able to:

- Explain the inputs / outputs of an Electrolyzer system
- Discuss how to calculate losses and equipment efficiency

Explain

The inputs to an electrolyzer are:

- DC electric power supply (from renewable sources of energy)
- Water
- Electrolyte (in case of alkaline electrolyzer)

Explain the various precautions they should keep in mind when providing inputs to electrolyzer.
The output from electrolyzer is hydrogen and the by product is oxygen.

Explain to them that extreme care should be taken when handling hydrogen.

Explain the process of calculating losses and equipment efficiency.

- The efficiency tests are performed in authorized laboratory.
- Once the electrolyzer passes the test, test certificates are issued.
- Test performed on individual equipment include:
  - Efficiency
  - Pressure test
  - Temperature withstand test
- Explain how plant efficiency and quantity of hydrogen produced is calculated.

Ask

- Ask the participant if they know how to calculate the efficiency of an electrolyzer.

Notes for Facilitation

- Tell them about the equation for the water electrolysis reaction.
- Explain how to derive the efficiency of an electrolysis system.
- Tell them about HHV— the higher heating value of hydrogen.
- Illustrate with the help of the schematic diagram that shows the components of the PEM electrolyzer plant.

UNIT 4.4: Input Renewable Power from various sources and its Integration with Electrolyzer

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the various renewable sources of energy and their limitations
2. Explain the concept of integration of incoming renewable power for supply to the electrolyzer

Explain

Tell the learner about the different types of renewable energy sources and their limitations.

Explain the concept of Solar Energy, Wind Energy, Hydro Energy.

Explain the requirements for Integration of Renewable energy with Electrolyzer.

Explain Integration of solar PV, Wind Energy, Hybrid wind and solar power for green hydrogen production and usage.

Ask

You may question the learners if they have any idea about the integration of renewable energy with green hydrogen plant in terms of the activities needed to be performed.

Notes for Facilitation

Tell the learners about the requirements and available alternatives for integration of Renewable Energy.

- Requirement: The electrolyzers operate on DC at low voltage and high current.
- Available alternatives:

(i) DC electric power from Solar PV: D.C. electric power is fed to DC-DC converter to convert DC voltage to match with the DC voltage of electrolyzer.

(ii) AC electric power from wind and hydro: AC electric power is converted to DC electric power by rectifier. A DC-DC converter changes DC voltage to match with the DC voltage of electrolyzer.

(iii) Hybrid e.g. DC power from solar PV and AC power of wind or hydro: The AC electric power output of wind and/or hydroelectric power projects is converted to DC by rectifier. Next, DC-DC converter changes DC voltage to match with the DC voltage of electrolyzer.

UNIT 4.5: Assembly/Installation of Parts and Components of Electrolyzer

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the assembly/Installation of Parts and Components of Electrolyzer.
2. Discuss step by step process for assembly/Installation of Electrolyzer- Bipolar plate, separator (membrane), Porous Transport layer, electrodes etc.

Ask

Ask the participants if they know how to assemble an electrolyzer.

Say

- The electrolyzers supplied by the factories are in a pre-assembled form.
- These units can then be assembled at the plant as per the intended production capacity.
- However, for small plants, it is more cost effective to acquire factory assembled electrolyzers as assembling an electrolyzer requires:
  - Expertise
  - Specific tools and equipment
  - Meticulous testing procedures
- The assembling process should be carried out in a controlled environment to ensure:
  - Proper sealing
  - Membrane integrity
  - Precise oxygen and hydrogen outlets

Notes for Facilitation

- Introduce the topic of assembling of an electrolyzer.
- List the prerequisites for assembling an electrolyzer.
- Tell them about the importance of carrying out the assembling process in controlled environment at the plant.
- Explain the assembling process in detail.
- Illustrate with the help of diagrams and flowchart of the process.

Activity

- Divide the participants into pairs.
- Ask them to show how to handle different tools and equipment as per concerned standard and industry practices.

Demonstrate

The stack assembly of a PEM electrolyzer.

Notes for Facilitation

- Explain the step-by-step process of assembling a PEM electrolyzer.
- Show them how to assemble a PEM electrolyzer.
- While demonstrating, tell them about the various checks they should perform at each step to ensure that the electrolyzer is assembled as per the guidelines.

UNIT 4.6: Challenges Associated with Hydrogen in Storage, Handling and Transportation

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the Challenges associated with Hydrogen in storage, handling and transportation

Explain

Explain the need for Hydrogen storage. Explain the figure showing Hydrogen storage and distribution.

Tell the learners about challenges in storing hydrogen.

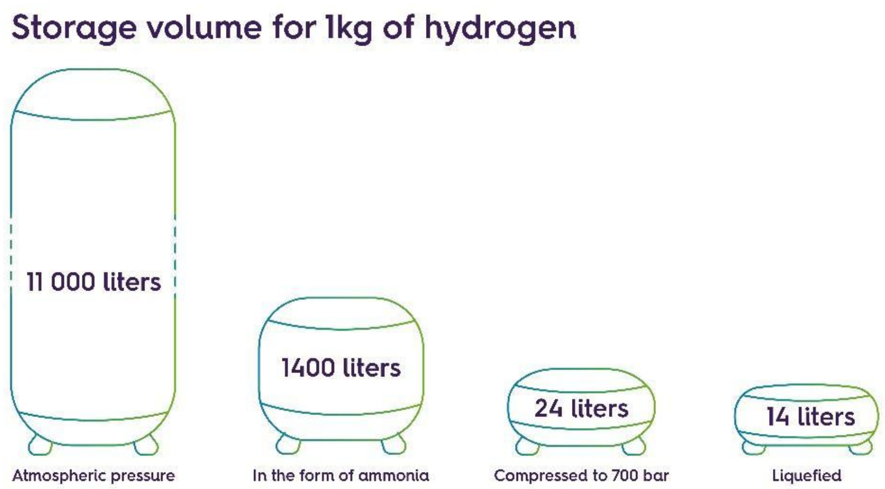
Show them the effect of pressure on the volume of hydrogen. Compare different modes of transportation with respect to the distance and quantity of hydrogen to be transported.

Ask

You may question the learners if they have any idea about the challenges associated with hydrogen storage and distribution in terms of the activities needed to be performed.

Notes for Facilitation

Tell the learner about the effect of pressure on the volume of hydrogen



UNIT 4.7: Hydrogen Compression System

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss how to select and install hydrogen compression and storage system
2. Discuss the precautions required to compress and store hydrogen
3. Discuss the safety guidelines to be followed as per industry standard

Explain

Explain the technical considerations that need to be considered in the selection of a hydrogen compression system.

Explain the table given on the slide that gives data on compressors in NTPC projects for reference

Tell the learners about Parts of a centrifugal compressor.

Do

You may show a small video on how to maintain the suction and discharge valves in a compressor using the following link:

https://www.youtube.com/watch?v=knFiyPZqM1M

Notes for Facilitation

Tell the learner about Key components/parameters to be considered.

- The Factor of safety more than 2 i.e., all the components should withstand double of storage pressure.
- Change of seals after certain hours of storage or operation under pressure.
- Valves are the heart of a compressor. The performance of a reciprocating air compressor mostly depends on a suction filter, valves, and piston rings. Hence due care is required to be taken. The valves require periodic inspection and maintenance.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 4.8 O&M Requirements for Electrolyzer

Unit Objectives

At the end of this unit, you will be able to:

1. Explain key O&M requirements for electrolyzer

Ask

- Ask the participant if they know how to perform operation and maintenance (O&M) activities for electrolyzer.

Demonstrate

- Show the participant how to perform operation and maintenance (O&M) activities for an electrolyzer.

Notes for Facilitation

- Tell them about:
  - The tasks involved in operation activities
  - The tasks involved in maintenance activities
- Tell them about the importance of maintaining extreme care when doing maintenance activities.
- Tell them that the best practices for handling water electrolysis cell stacks comprise:
  - Pre-check
  - Safety measures
  - Voltage verification
  - Additional safeguard
- Explain each step in detail.
- After demonstrating the O&M procedures, ask the participants to share their queries.

Demonstrate

Show the participant how to

- Perform cleaning of an electrolyzer
- Perform checking of instruments

Explain

The steps involved in cleaning of an electrolyzer.

- Cleaning filter mesh
- Maintenance of compressor
- Maintenance of dryer
- Recharging of hydrogen

There are a lot of tools that are used in a green hydrogen production plant. All these tools need to be cleaned thoroughly and regularly. Failure to do so may result in inaccurate results, faults and defects in equipment.

Check:

- Zero position of instruments
- Calibration of instruments
- Removal of oil from instruments

The start-up sequence of an electrolyzer:

- Turn on the main switches
- Start DM feed
- Start electrolyte pump
- Set temperature
- Check purity of hydrogen after the plant has been working for few hours

Routine operation and maintenance must be performed after the plant has been operational for few hours to ensure that all the parts are working properly and that there are no anomalies.

Demonstrate

Show the participant how to

- Perform the startup sequence
- Perform routine maintenance

Notes for Facilitation

- Tell them about the importance of cleaning the electrolyzer.
- After demonstrating the steps of cleaning, ask the participants to share any queries they might have.
- Next, tell them about the correct way to clean the instruments.
- Show them how to properly clean the instruments.
- List the steps of the startup sequence.
- After demonstrating the startup sequence:
    - Group the participants into pairs
    - ask them to perform the startup sequence

Ask

- Ask the participant if they know some safety precautions that must be observed while performing maintenance tasks.

Demonstrate

The safety precautions to be observed while doing maintenance tasks.

Notes for Facilitation

- Tell them about the importance of observing safety precautions when doing maintenance activities.
- List the safety precautions, such as:
  - Strictly following the operating rulea and regulations
  - Installing explosion -proof lights
  - Not carrying out any repairs when the plant is running
  - No smoking
  - Easy access to fire-proof equipment
  - Use of PPE
  - Keeping the surface of electrolyzer clean
  - Remedial measures to be taken in case of any accident
  - Restricting entry to room storing explosive materials.

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the participants to list some safety precautions that must be observed when operating the plant and doing maintenance activities.
- Ask the group to discuss and write down the answers.
- Give the groups 10 minutes for the activity.

5. Micro-entrepreneurship opportunities in Green

SGJ/N1819

Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss micro-entrepreneurship opportunities for
2. Uninterrupted quality water supply for electrolyzer, water quality check
3. Installation and commissioning of electrolyzer
4. Renewable energy power supply arrangement plant for green hydrogen plant
5. Storage of green hydrogen
6. Installation of pipelines in green hydrogen plant/industry
7. Safe transportation of green hydrogen to consumers
8. Handling of salt enriched water from water feed input system, if required, depending on the type of electrolyser
9. Managing safety of the plant
10. Operation and Maintenance of green hydrogen plant
11. Leakage monitoring, purging of hydrogen by CO2/NH3 before start of work. Recharging of hydrogen by avoiding air contact by using again CO2/NH3 and then filling of hydrogen

UNIT 5.1: Micro-entrepreneurship opportunities for water supply, installation and commissioning of electrolyzer

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship opportunities for Uninterrupted quality water supply for electrolyzer, water quality check
2. Discuss micro-entrepreneurship opportunities for Handling of salt enriched water from water feed input system, if required, depending on the type of electrolyser
3. Discuss micro-entrepreneurship opportunities for Installation and commissioning of electrolyzer

Explain

Explain Micro-entrepreneurship opportunities for Quality Water Supply.

Tell the learner about Micro-Entrepreneurship Opportunities for Handling of salt enriched water from water feed input system.
Explain Micro-Entrepreneurship Opportunities for Installation and commissioning of electrolyzer.

Ask

You may question the learners if they have any idea about Micro-entrepreneurship Opportunities for Water Supply, Installation and Commissioning of Electrolyzer.

Notes for Facilitation

Tell them about Micro-entrepreneurship opportunities for Quality Water Supply given below.

Water Supply Solutions
Water Quality Monitoring
Water Treatment Solutions
Mobile Water Testing Labs
Water Supply Chain Management
Consultancy and Training
Remote Sensing and IoT Solutions
Water Recycling and Circular Solutions

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to write down Micro-entrepreneurship Opportunities for Water Supply, Installation and Commissioning of Electrolyzer.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

UNIT 5.2 Micro-entrepreneurship Opportunities for Renewable Energy Power Supply

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship opportunities for Renewable energy power supply arrangement for green hydrogen plant.

Explain

Recap the main source of renewable energy in India.
Explain the need for integration of renewable energy.
List the limitations of each type of renewable resource.
Explain, with the help of chart Micro-Entrepreneurship Opportunities for Renewable Energy Supply Arrangement Plant for Green Hydrogen Plant.

Notes for Facilitation

Tell the learners about Micro-Entrepreneurship Opportunities for Renewable Energy Power Supply Arrangement Plant for Green Hydrogen Plant.

- Solar Power Solutions
- Wind Energy Services
- Hybrid Energy Systems
- Microgrid Development
- Energy Storage Solutions
- Energy Management and Optimization
- Off-Grid Solutions
- Maintenance and Repairs
- Energy Audits and Consultancy
- Financing and Leasing Options

UNIT 5.3 Micro-entrepreneurship Opportunities for Storage, Transportation, and Installation

Unit Objectives

At the end of this unit, you will be able to:

- Discuss micro-entrepreneurship opportunities for Storage of green hydrogen
- Discuss micro-entrepreneurship opportunities for Safe transportation of green hydrogen to consumers
- Discuss micro-entrepreneurship opportunities for Installation of pipelines in green hydrogen plant/industry

Explain

Explain the Micro-entrepreneurship opportunities related to the storage of green hydrogen.
Tell the learners about Micro-entrepreneurship opportunities related to the safe transportation of green hydrogen to consumers.
Explain the Micro-entrepreneurship opportunities related to the installation of pipelines in green hydrogen plants and industries

Do

You may show a small video on the Business Opportunities in Green Hydrogen Mission of India using the following link:

https://www.youtube.com/watch?v=rHo7rJSvVfA

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to write down Micro-entrepreneurship Opportunities for Storage, Transportation, and Installation.
- Ask the group to discuss and write down the answers.
- Give the groups 5 minutes for the activity.

Say

Summarize the topics taught in the module. You may ask the learners to recall what all has been taught.

UNIT 5.4 Micro-entrepreneurship for Safety, Operation and Maintenance of Green Hydrogen Plant

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship opportunities for managing safety of the plant
2. Discuss micro-entrepreneurship opportunities for Operation and Maintenance of green hydrogen plant.

Explain

Explain the Micro-entrepreneurship Opportunities for Managing Safety of the Plant
Tell the learners about Micro-entrepreneurship Opportunities for Operation and Maintenance of Green Hydrogen Plant.

Notes for Facilitation

Tell the learner about key micro-entrepreneurship opportunities for Operation and Maintenance of Green Hydrogen Plant:

- Plant Operation Services: Entrepreneurs can offer plant operation services, including the operation of hydrogen production equipment, monitoring of processes, and ensuring continuous production.
- Maintenance and Repairs: Micro-entrepreneurs can specialize in maintenance and repair services for hydrogen production equipment, ensuring minimal downtime and optimal performance.
- Condition Monitoring Solutions: Entrepreneurs can develop and provide condition monitoring solutions that use sensors and predictive maintenance techniques to detect potential equipment failures before they occur.
- Plant Optimization Services: Micro-entrepreneurs can focus on optimizing plant operations by improving energy efficiency, reducing waste, and enhancing overall performance.
- Remote Monitoring and Control: Entrepreneurs can offer remote monitoring and control systems that allow plant operators to manage equipment and processes from a centralized location, improving efficiency and safety.
- Hydrogen Quality Assurance: Micro-entrepreneurs can provide services for monitoring and ensuring the quality of the produced hydrogen, including purity levels and moisture content.
- Plant Safety Inspections: Entrepreneurs can offer safety inspection services to assess and enhance the safety protocols and practices within the green hydrogen plant.

UNIT 5.5 Micro-entrepreneurship for Leakage Monitoring

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss micro-entrepreneurship Opportunities for Leakage Monitoring
2. Discuss micro-entrepreneurship opportunities for purging of hydrogen by CO_2/NH_3 before start of work.
3. Discuss micro-entrepreneurship opportunities for Recharging of hydrogen by avoiding air contact by using again CO_2/NH_3 and then filling of hydrogen.

Explain

Explain the Micro-entrepreneurship Opportunities for Leakage Monitoring.
Tell the learners about micro-entrepreneurship opportunities for purging of hydrogen by CO_2/NH_3 before start of work.
Explain the Micro-Entrepreneurship Opportunities for Recharging of hydrogen by avoiding air contact by using again CO_2/NH_3 and then filling of hydrogen

Team Activity

- Divide the class into five to six groups of four participants each.
- Give blank paper and pens to each group.
- Ask the groups to write down Micro-entrepreneurship for Leakage Monitoring.
- Ask the group to discuss and write down the answers.

Give the groups 5 minutes for the activity

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the Micro-entrepreneurship and others would ask questions.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught

Skill India
कौशल भारत - कुशल भारत

सत्यमेव जयते
GOVERNMENT OF INDIA
MINISTRY OF SKILL DEVELOPMENT & ENTREPRENEURSHIP

N·S·D·C
National Skill Development Corporation
Transforming the skill landscape

SCGJ | SKILL COUNCIL FOR GREEN JOBS

6. Perform Health and safety measures for installing and operating green hydrogen systems

SGJ/N0802

Key Learning Outcomes

At the end of this module, you will be able to:

1. Explain the requirements for safe work area at hydrogen production project site.
2. Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems
3. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.
4. Discuss importance of different detectors and safety tools
5. Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures
6. Explain need to maintain ideal temperature and humidity level of storage areas used to safely contain gas cylinders.
7. Utilize sensors that can alert the responsible person such as a safety officer when storage rooms are not maintaining the ideal conditions for storing hazardous chemicals.
8. Explain the importance of administering first aid.
9. Identify the personal protective equipment used for the specific purpose.
10. Identify the hazards associated with hydrogen production system;
11. Identify work safety procedures and instructions for working at hydrogen production plant complying with applicable safety regulations
12. Discuss all applicable statutory requirements along with safety regulations in terms of fire protection.
13. Incorporate good housekeeping practices and infection control guidelines.

UNIT 6.1: Occupational Health & Safety Standards and regulations

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the requirements for safe work area at hydrogen generation project site.
2. Explain the importance of Occupational health & Safety standards and regulations for Basic considerations for the safety of hydrogen systems

Explain

- Start by telling them that the occupational health and safety standards and regulations for hydrogen production plants are designed to protect workers from the hazards associated with hydrogen production and handling.
- Explain to them about fuels and their degree of hazard.
- Tell them about some guidelines to enable the safe handling and use of fuel.
- Tell them about major Caution like Finding leaks and having adequate ventilation are essential. Since hydrogen burns with a flame that is almost invisible, specialized flame detectors and thermo-vision cameras are needed.
- Tell them about the health effects caused by hydrogen:

Asphyxiation	Headache	Dizziness	Nausea	Vomiting	Frostbite

Fig 6.1.1: Health effect caused by hydrogen

- Explain that OSHA stands for Occupational Safety and Health Administration and its mission is to "assure safe and healthful working conditions for workers by setting and enforcing standards and by providing training, outreach, and education."
- Tell them that IOS stands for International Organization for Standardization.
- The ISO standards that apply to hydrogen production plants include:

ISO 14687
- This standard specifies the safety requirements for hydrogen production, storage, and transportation.

ISO 15916
- This standard specifies the requirements for the design, construction, and operation of hydrogen fuel cell systems.

ISO 22719
- This standard specifies the requirements for the design, construction, and operation of hydrogen refueling stations.

Fig 6.1.2: IOS standard

UNIT 6.2: Emergency Situations in a Hydrogen System

Unit Objectives

At the end of this unit, you will be able to:

1. List the probable emergency situations in a hydrogen system.
2. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.

Ask

Ask the participants about the colour of the flame of hydrogen gas.

Notes for Facilitation

- Tell them about the Hydrogen Flammability and explosion.
- Tell them about Hydrogen Ignition Energy.
- Tell them about the natural calamities.

Explain

- Explain to them that Hydrogen is a highly flammable gas with a wide flammability range of 4% to 75% in air. It means that a hydrogen-air mixture can ignite even with a small amount of hydrogen present.
- Tell them that the Hydrogen also has a very low ignition energy, meaning that it can be ignited by a spark or even static electricity.
- Tell them that natural calamities are the events that are caused by natural forces and can cause widespread damage and loss of life.
- Tell them about some safety precautions which should be taken to prevent hydrogen explosion. These precautions include:

Avoiding ignition sources, such as sparks, flames, and static electricity.	Ventilating areas where hydrogen is present.	Using hydrogen detectors to monitor for leaks.	Storing hydrogen in safe containers.

Fig 6.2.1: Safety precautions

- Some of the most common natural calamities include:

Earthquakes	Floods	Hurricanes	Tornadoes	Wildfires

Fig 6.2.2: Some natural calamities

UNIT 6.3: Detectors and Safety Tools

Unit Objectives

At the end of this unit, you will be able to:

1. Discuss the importance of detectors and safety tools
2. Describe potential causes of emergency such as gas leaks, fire, explosion, bomb threatening, natural calamities etc.

Ask

- Ask the participants about the importance of leak detection of hydrogen gas.
- Ask the participants about the safety majors of hydrogen explosion.

Notes for Facilitation

- Tell them about the methods of leak detection.

Explain

- Tell them about the different methods for detecting hydrogen leaks, such as:

  - Portable hydrogen detectors: It is used to detect hydrogen leaks in the air.
  - Fixed hydrogen detectors: They are permanently installed in areas where hydrogen is present.
  - Bubble testing:It is performed by simply applying a soapy solution. If there is a leak, bubbles will form.
  - Infrared cameras: It detects hydrogen leaks by detecting the heat signature of the escaping gas.
  - Acoustic leak detection: It uses microphones to listen for the sound of hydrogen escaping.

Fig 6.3.1: Methods of leak detection

- Tell them about Indicative shutdown control actions for hydrogen systems are actions these are taken to safely shut down a hydrogen system in the event of a leak or any other emergency.

UNIT 6.4: Material Safety

Unit Objectives

At the end of this unit, you will be able to:

1. Review the Material Safety Data Sheet and labels of chemicals contained in cylinders in order to be aware of their hazards and precautionary measures.

Notes for Facilitation

- Tell them about Globally Harmonized System (GHS).
- Tell them about Material Safety Data Sheet (MSDS),

Explain

- Start by telling them about the Globally Harmonized System (GHS) of Classification and Labelling of Chemicals
    - It is a system for classifying and labelling chemicals so they can be handled, utilized, and transported safely.
    - Under the direction of the United Nations Economic and Social Council, the GHS was established by the TDG, or United Nations Sub-Committee of Experts on the Transport of Dangerous Goods.
    - More than 70 nations, including the US, Canada, and the EU, have endorsed and implemented the GHS since it was adopted by the UN in 2003.
- Tell them about the Material Safety Data Sheet (MSDS), also known as a Safety Data Sheet (SDS).
    - MSDSs can be used to develop and implement safe handling procedures for hazardous chemicals.
    - For example, the information in an MSDS can be used to:

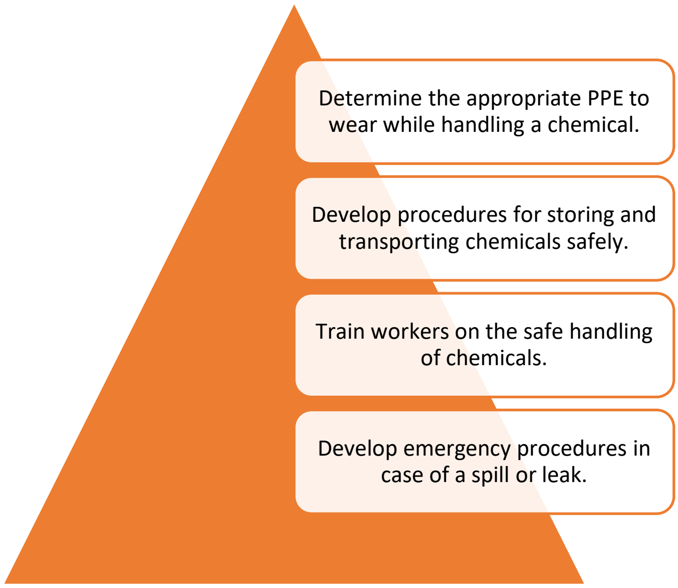


Fig 6.4.1: Uses of MSDS

UNIT 6.5: Maintain Ideal Temperature and Humidity Level

Unit Objectives

At the end of this unit, you will be able to:

1. Explain the need to maintain ideal temperature and humidity to safely contain hydrogen gas cylinders.

Notes for Facilitation

- Tell them about the effects of temperature and humidity on hydrogen production and purity.

Explain

- Explain that the change in temperature and humidity can have a significant effect on the purity of hydrogen gas.

Effects/changing conditions	Temperature(increasing)	Humidity
Effect on hydrogen purity	It increases the amount of impurities present	Increasing the humidity of hydrogen gas can decrease its purity.
Effect on hydrogen production	It also increases the purity of the hydrogen gas produced.	Increasing the humidity of the hydrogen production process can also decrease the purity of the hydrogen gas produced.

Table 6.5.1: Change in temperature and humidity can affect the purity of hydrogen gas

- Explain to them about the importance of monitoring temperature and humidity.
- Temperature and humidity can affect the comfort and safety of people, as well as the quality of products and equipment, so it is important to monitor it.

Thermometers and hygrometers	Data loggers	Building management systems (BMS)

Fig 6.5.1: Different devices to monitor temperature and humidity

UNIT 6.6: Smart Hazardous Chemical Storage Monitoring

Unit Objectives

At the end of this unit, you will be able to:

1. Utilize sensors that can alert the responsible person such as a safety officer when storage rooms are not maintaining the ideal conditions for storing hazardous chemicals.

Ask

- Ask them the need to monitor the hazardous chemical storage.

Notes for Facilitation

- Tell the participants about the smart hazardous chemical storage monitoring system.
- Explain the uses and benefits of smart hazardous chemical storage monitoring system.

Explain

- Explain to them the uses of smart hazardous chemical storage monitoring systems.
- Tell the participants about the key components of a smart hazardous chemical storage monitoring system:

Sensors → Data Analytics → Communication Technologies

Fig 6.6.1: Key component of smart hazardous chemical storage monitoring system

- Tell them about the applications of smart monitoring, such as:

Leak Detection
Temperature Monitoring
Inventory Management
Access Control
Environmental Monitoring

Fig 6.6.2: Application of smart hazardous chemical storage monitoring system

- Tell the participants about the benefits of smart hazardous chemical storage monitoring system.

UNIT 6.7: Administering First Aid

Unit Objectives

At the end of this unit, you will be able to:

1. Know the importance of first aid and method of its administering.

Notes for Facilitation

- Tell the participants about the first aid during the injuries and exposures caused due to hydrogen.

Explain

- Tell them about the health conditions in which an employee should work or should not work.
- Explain to them the importance of maintaining strict adherence to safety protocols and health precautions due to the flammable and potentially hazardous nature of hydrogen gas.
- Tell them about some of the essential health precautions to take while working at a hydrogen plant:

Fig 6.7.1: Essential health precautions

Check for Consciousness: Check for breathing and a pulse. If necessary, initiate CPR (cardiopulmonary resuscitation) immediately.

Hydrogen Gas Exposure: If the victim has been exposed to hydrogen gas, remove them from the contaminated area to fresh airIf breathing is difficult, administer oxygen.

Eye Exposure: Fush the eyes with clean water for at least 15 minutes, holding the eyelids open. Seek medical attention immediately.

Skin Exposure: Remove contaminated clothing and wash the affected area with soap and water for at least 15 minutes. If irritation or burns persist, seek medical attention.

Ingestion: If the victim do not induce vomiting. Seek immediate medical attention.

Monitor and Reassure: Monitor the victim's vital signs and keep them warm and comfortable.

Emergency Medical Services: Call for emergency medical services as soon as possible.

Follow-up Care: Ensure that the victim receives appropriate medical attention.

Fig 6.7.2: First Aid during injuries

UNIT 6.8: Personal Protective Equipment

Unit Objectives

At the end of this unit, you will be able to:

1. Identify the personal protective equipment used for the specific purpose.

Ask

- Ask the participants about the use of Personal Protective Equipment.

Notes for Facilitation

- Tell the participants about the PPE (personal protective equipment).

Explain

- Tell the participants about Personal protective equipment (PPE), which is designed to protect workers from workplace hazards that could cause injury or illness.
- The type of PPE required depends on the specific hazards present in the work environment.
- Tell them about the safety it provides for the user.
- Common types of PPE include:

Fig 6.8.1: Common types of PPE

UNIT 6.9: Hazards Associated with Hydrogen production System

Unit Objectives

At the end of this unit, you will be able to:

1. Know hazards associated with Hydrogen Production System.

Notes for Facilitation

- Tell the participants about the hydrogen production equipment.
- Tell them about the spillage and leakage hazards.

Explain

- Explain to them that the Hydrogen production equipment can pose various health hazards due to the flammable and reactive nature of hydrogen gas. These hazards can lead to fires, explosions, and other safety incidents if not properly managed.
- Tell them about some of the primary hazards associated with hydrogen production equipment, such as:

Flammability and Explosivity

Low Ignition Energy

Invisibility and Odorlessness

High Pressure and Cryogenic Temperatures

Material Compatibility

Asphyxiation Hazard

Fig 6.9.1: Hazards associated with Hydrogen production

- Explain to them about the spillage and leakage hazards as these incidents can lead to fires, explosions, and other safety concerns if not properly managed.

Fire and Explosion Hazards

Invisibility and Odorlessness

Asphyxiation Hazard

Environmental Hazards

Health Hazards

Fig 6.9.2: Hazards caused by spillage and leakage

- Tell them about various safety measures.

UNIT 6.10: Work Safety Procedures and Instructions

Unit Objectives

At the end of this unit, you will be able to:

1. Know safety procedures to be followed in hydrogen production plant.

Notes for Facilitation

- Tell the participants about the Guide to Safety of Hydrogen and Hydrogen Systems.
- Tell them about the storage, labelling before transportation of Hydrogen.

Explain

- Explain to them Guide to Safety of Hydrogen and Hydrogen Systems. Explain about Safety Standard for Hydrogen and Hydrogen Systems used by NASA and also the Sourcebook for Hydrogen Applications
- Tell them about storage, handling and labelling before the transportation of hydrogen.
- Explain about Pre-departure Checks for trucks Carrying Hydrogen in Cylinders in Manifold Systems.
- Tell the learner about Labels and Markings for transportation.
- Explain the Precautions during transportation.

Role Play

Educate them on how to ask the previously explained pertinent questions.

Then do a role play, where one learner would be the trainer and others would ask questions.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 6.11: Fire Protection

Unit Objectives

At the end of this unit, you will be able to:

1. Understand the applicable statutory requirements and safety regulations for fire protection

Explain

- Explain various standards related to fire on Hydrogen issued by NFPA.
- Tell them about best practices for dealing with a gaseous hydrogen fire.

Notes for Facilitation

- Tell the participants about the standards on Hydrogen.

Name of Standard	**Short Title**
NFPA 2	Hydrogen Technologies Code: This code provides fundamental safeguards for the generation, installation, storage, piping, use, and handling of hydrogen in compressed gas (GH2) form or cryogenic liquid (LH2) form
NFPA 30A	Rules for design of refueling stations
NFPA 50A	Standard for gaseous hydrogen systems at consumer sites
NFPA 50B	Standard for liquefied hydrogen systems at consumer sites
NFPA 55	Compressed gases and cryogenic fluids code
NFPA 70	National Electrical Code: It is the benchmark for safe electrical design, installation, and inspection to protect people and property from electrical hazards. These can be used for selection of the electrical fittings in hydrogen handling areas.

Summarize

Summarize the topics covered in the module.

You may ask the learners to recall what all has been taught.

UNIT 6.12: Housekeeping and Infection Control

Unit Objectives

At the end of this unit, you will be able to:

1. Realize the importance of good housekeeping and infection control.

Explain

- Explain the importance of Housekeeping.
- Tell them about Good Housekeeping practices. Explain principles of Good Housekeeping.
- Explain the precautions to be taken for protection against infectious diseases.

Demonstrate

Demonstrate good housekeeping and infection control & prevention practices.

Notes for Facilitation

- Tell the participants about the benefits of housekeeping and cleanliness.

Summarize

Summarize the topics covered in the module.
You may ask the learners to recall what all has been taught.

Employability Skills

It is recommended that all trainings include the appropriate Employability skills Module.

Content for the same can be accessed

Employability skills workbook

Scan/ Click QR Code to access

Annexure

Module No.	Unit No.	Topic Name	Page no.	URL	QR Code(s)
Module 1 - Introduction to Green Hydrogen	Unit 1.1	What is Green hydrogen	5	https://www.youtube.com/watch?v=GqVxhE2cVGk	
Module 1 - Introduction to Green Hydrogen	Unit 1.3	How to produce hydrogen gas at home	16	https://www.youtube.com/watch?v=RAbjgk1Zlwc	
Module 1 - Introduction to Green Hydrogen	Unit 1.6	Hydrogen Electrolyzer	28	https://www.youtube.com/watch?v=WfkNf7kMZPA	
Module 1 - Introduction to Green Hydrogen	Unit 1.6	What is Hydrogen Fuel with Full Information	28	https://www.youtube.com/watch?v=Bn6_fQGqiJQ	
Module 1 - Introduction to Green Hydrogen	Unit 1.7	Renewable Energy Sources	32	https://www.youtube.com/watch?v=wl6eKPZ_qkc	
Module 2 - Components of Green Hydrogen Plant and its Layout	Unit 2.1	Demineralization process	47	https://www.youtube.com/watch?v=gCMxFmGkG8s	
Module 3 - Key technical and entrepreneurial aspects for supporting growth and business development green hydrogen production	Unit 3.1	How to Prepare a Bankable Project Report	113	https://www.youtube.com/watch?v=Ke_jwp_lskg	

Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.1	PEM Electrolyzer	135	https://www.youtube.com/watch?v=IX93qZbaiuo	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.2	Gigawatt green hydrogen plant	140	https://www.youtube.com/watch?v=fw5zPUbPo1Q	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.5	Installation of a Sunfire Pressurized Alkaline Electrolyzer	152	https://www.youtube.com/watch?v=qsygJibJ7eQ	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.5	Fuel cell and dismantling	154	https://www.youtube.com/watch?v=ykDiIH7Gtv0	
Module 4 - Oversee the Assembly, storage and O&M of Electrolyzer for Green Hydrogen Production	Unit 4.7	How to maintain the suction and discharge valves	166	https://www.youtube.com/watch?v=knFiyPZqM1M	
Module 5 - Micro-entrepreneurship opportunities in Green Hydrogen	Unit 5.3	New Business Opportunities Under the National Green Hydrogen Mission	185	https://www.youtube.com/watch?v=rHo7rJSvVfA	
Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.1	Occupational Health & Safety Standards and Regulations	200	https://www.youtube.com/watch?v=H6wRRWi6i0c	

Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.3	Best Hydrogen Detector	206	https://www.forensicsdetectors.com/blogs/articles/hydrogen-gas-monitors-expert-guidance	
Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.7	ling hydrogen H2	215	https://www.youtube.com/watch?v=WeDAZLrk4do	
Module 6 - Perform Health and safety measures for installing and operating green hydrogen systems	Unit 6.8	National Safety Compliance: OSHA Safety Training (2017)	218	https://www.youtube.com/watch?v=YE-63aeECVs	

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